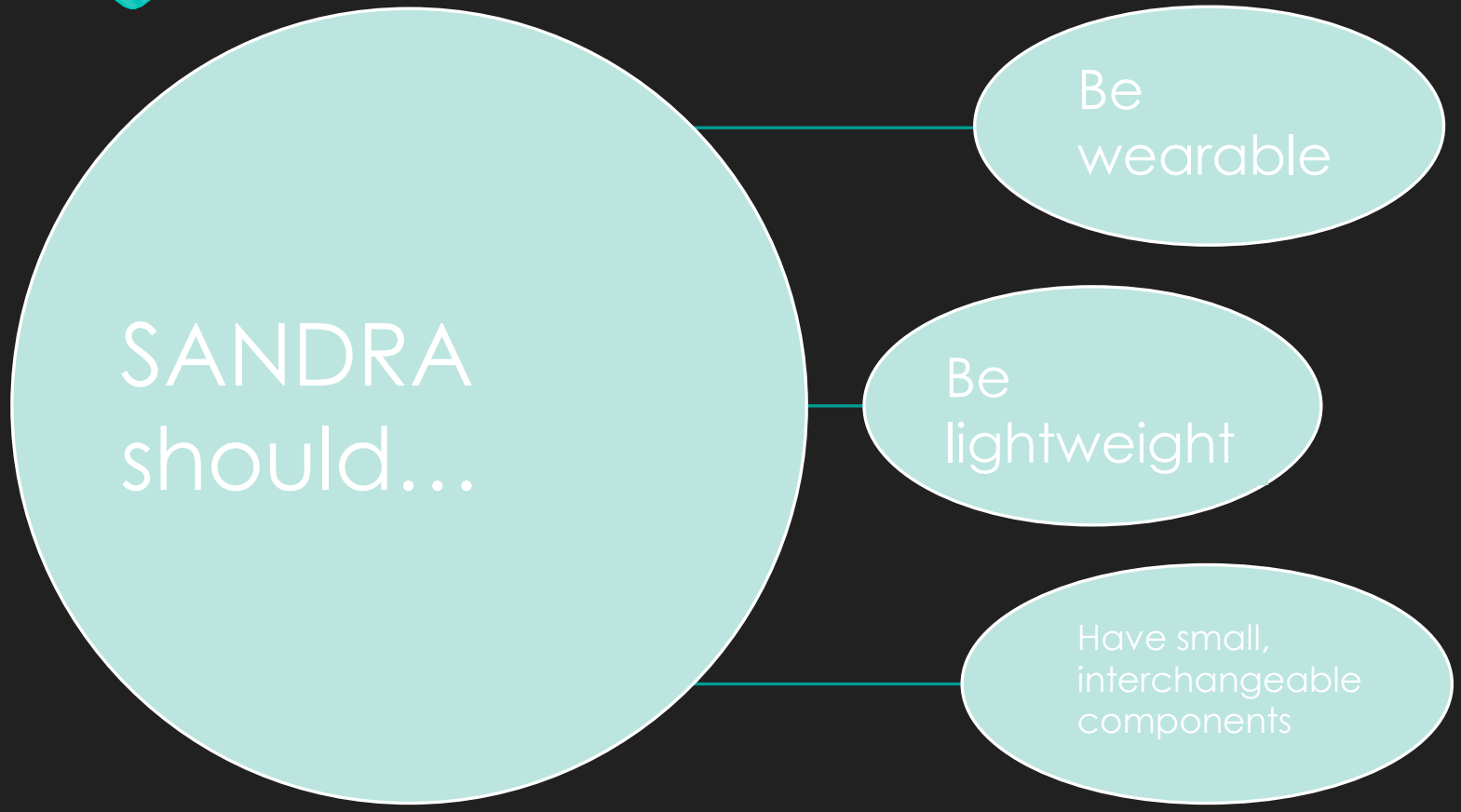
SANDRA (Serve and Data Results Analysis) Athletic Tech Glove

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Plano Independent School District
General Engineering II

Goal

Design and prototype a glove that collects telemetry in order to pinpoint areas of error in a volleyball serve

Initial Requirements



Introduction

Motivation

- Nothing self-contained, wearable, easily available, and environmentally versatile
- Many volleyball players struggle with their serve

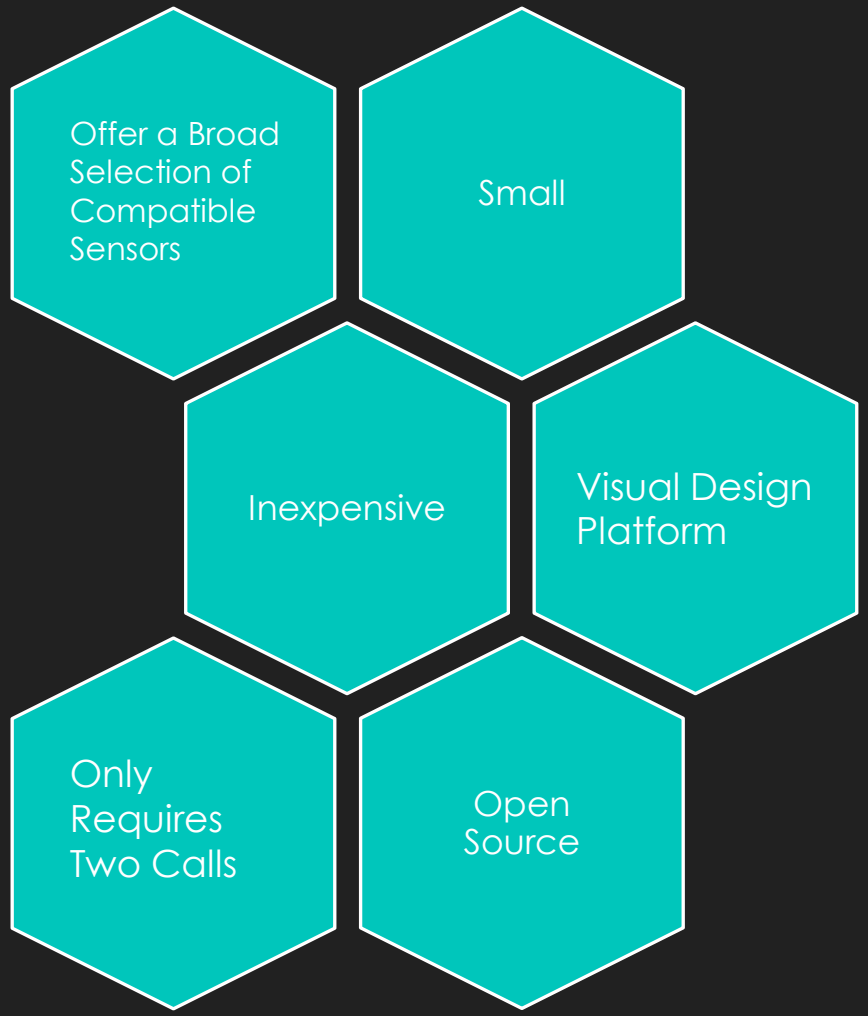
A volleyball serve must…

- Clear net both height- and distance-wise
- Overcome gravity

Microcontrollers…

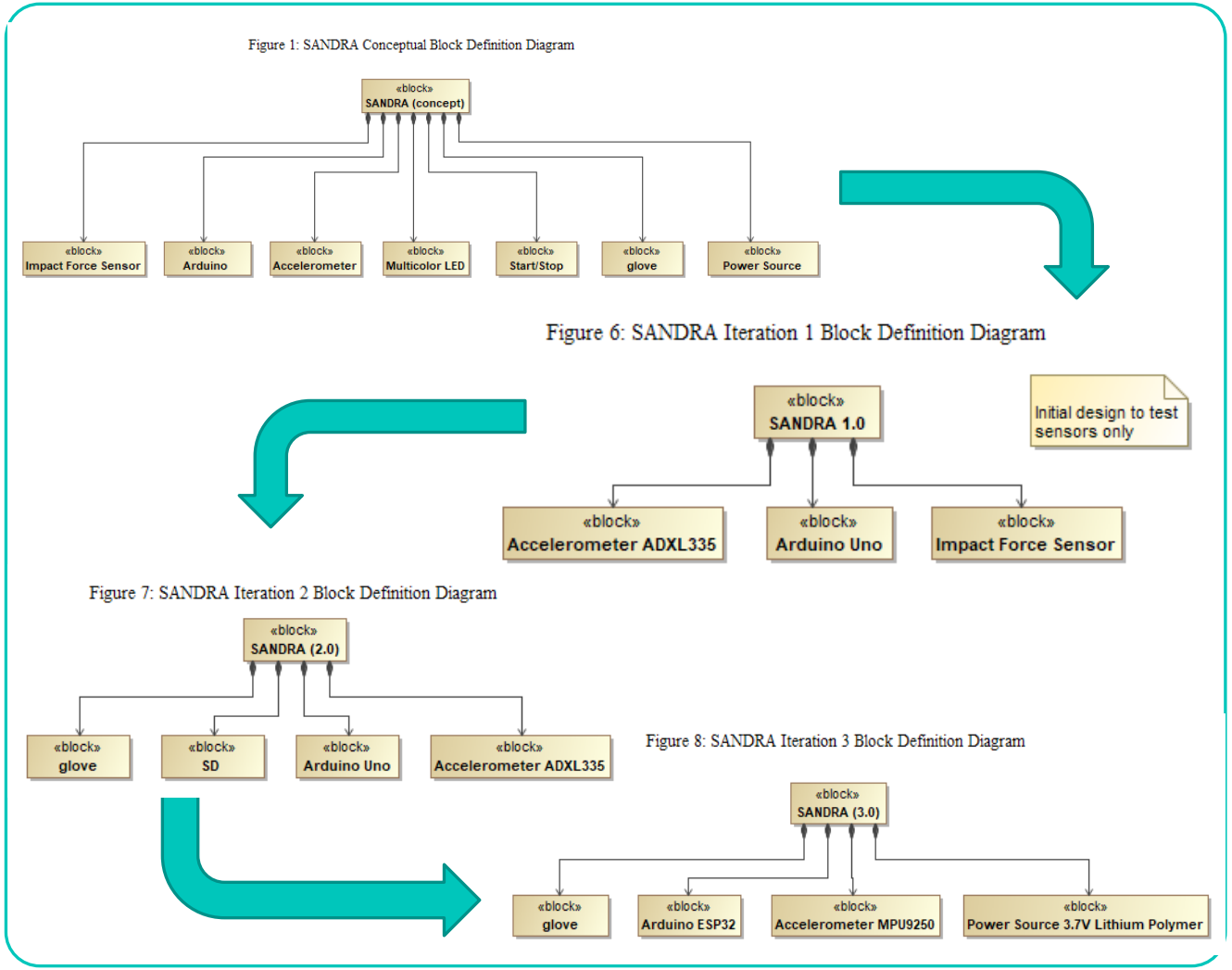
- are compact integrated circuits
- are designed to perform specific tasks
- typically combine a processor, read-only memory, and input-output functions

Why Arduino?


Offer a Broad Selection of Compatible Sensors
Small
Inexpensive
Visual Design Platform
Only Requires Two Calls
Open Source

From Concept to Implementation

- Conceptual: had multiple sensors and controls
- Iteration 1: used to verify sensor functionality
- Iteration 2: acceleration-to-angle feature; unnecessary components were removed
- Iteration 3: attempted to implement wireless connectivity; I^2C protocol



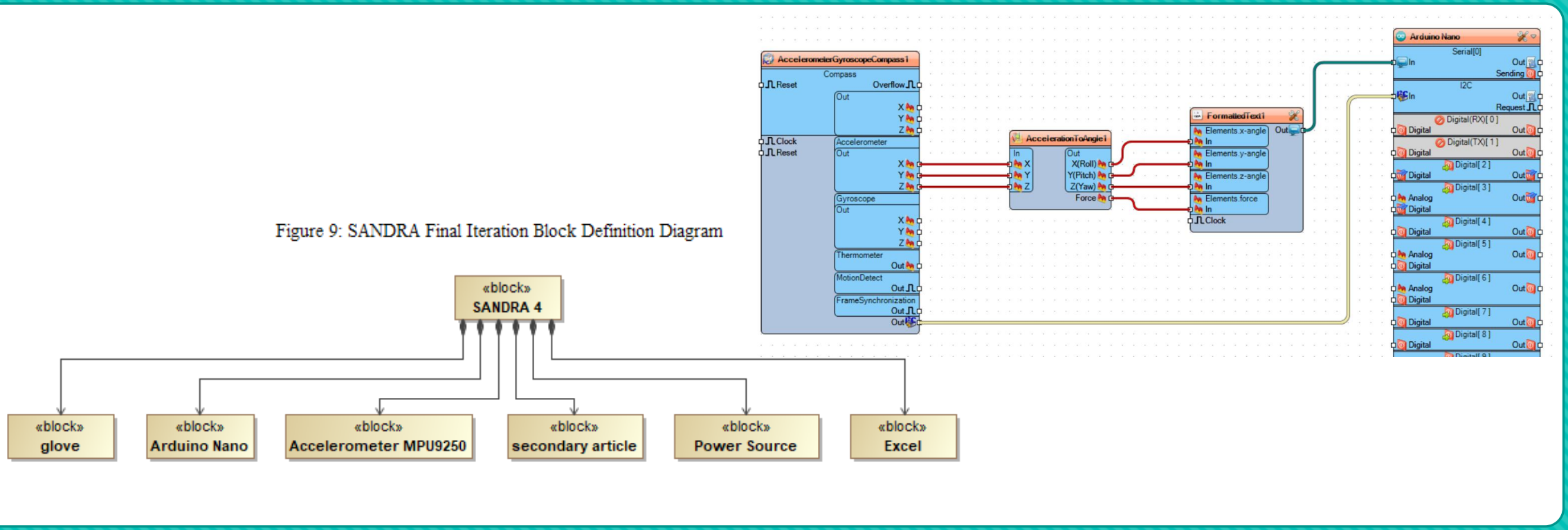


Figure 9: SANDRA Final Iteration Block Definition Diagram

Final Design

Materials

Electronics	Assembly
1 Arduino Nano ($22.00)	1 Right-handed Fingerless Glove ($3.00)
15 ft long cable- USB A to Mini ($2.19)	1 Athletic Wrist Sweatband ($3.50)
1 MPU9250 Accelerometer ($6.80)	¾" wide Velcro ($0.80)
4 Female to Female Dupont Wires ($0.20)	3 adhesives ($0.25)
Software: Visuino, MagicDraw, Excel	Foam Core Board ($0.07)
PC	

Total Production Cost*: $38.81



*for a single unit at the retail prices of the components

Sensor Calibration

Axis	Voltage @ 0g	Voltage @ 1g
X	270mV	400mV
Y	270mV	400mV
Z	270mV	400mV

- X-axis: front/back
- Y-axis: left/right
- Z-axis: up/down

- The sensor was calibrated by rotating it 90° along each axis, and voltage was captured at each stage.
- From this calibration, it was learned that 1g equates to 130mV.
- Angles and g-force are forms more meaningful for a SANDRA user.

Serve Calibration

The serve was calibrated by having a server, wearing SANDRA, perform 15 serves- both successful and unsuccessful- in order to baseline the data.

Trial	X-Angle (degrees)	Y-Angle (degrees)	Z-Angle (degrees)	G-Force (g-s)	Success
1	257.85	215.73	188.80	18.48	1
2	302.08	315.00	32.08	24.75	0
3	225.00	290.67	290.67	23.42	1
4	266.49	224.48	183.45	19.50	1
5	263.49	222.41	185.95	18.96	0
6	45.00	135.00	135.00	27.71	0
7	302.18	213.51	157.38	20.20	1
8	265.58	221.02	183.85	21.24	1
9	269.25	220.62	180.65	20.95	1
10	265.44	225.00	184.56	22.66	1
11	302.05	221.26	151.23	23.03	0
12	264.71	221.75	184.73	20.00	0
13	262.32	228.16	188.57	19.90	0
14	267.56	218.85	181.96	20.30	1
15	315.00	225.00	135.00	27.71	1

Unit	X	Y	Z	G-Force
Measurement	Above 100°	Between 200° and 291°	Above 157°	Above 16g-s

1. A failed serve requires either an orientation failure along two axes or insufficient G-Force.
2. A serve is more dependent on G-Force than on angle.

Serve Rules*

* Only for an underhand serve

Trials

- Test SANDRA's accuracy (specifically final three columns)
- Data analysis was performed using an Excel spreadsheet, which implements the rules for a successful serve using the conditional formatting function
- Validate by comparing to visual success
- The only inaccurately analyzed serve was Serve #11, which might be due to SANDRA's precision.
- SANDRA accurately reported the serve's status in **29/30 tests**, yielding an **accuracy of 96.7%**, which exceeded the threshold of 90% accuracy.

Trial	X-Angle (degrees)	Y-Angle (degrees)	Z-Angle (degrees)	G-Force (g-s)	Visual Success		Analysis		Accuracy	
1	280.82	204.29	175.07	16.57	✓	1	✓	1	✓	1
2	256.98	222.10	191.80	20.37	✓	1	✓	1	✓	1
3	349.24	190.90	134.61	15.02	✗	0	✗	0	✓	1
4	133.87	135.00	223.87	27.36	✗	0	✗	0	✓	1
5	296.79	303.45	37.39	20.81	✗	0	✗	0	✓	1
6	257.92	212.96	187.90	14.87	✗	0	✗	0	✓	1
7	315.00	225.00	135.00	27.71	✓	1	✓	1	✓	1
8	273.95	217.31	176.99	20.14	✓	1	✓	1	✓	1
9	323.66	209.98	141.90	22.33	✓	1	✓	1	✓	1
10	260.41	223.97	189.26	13.70	✗	0	✗	0	✓	1
11	266.59	214.26	182.33	17.08	✗	0	✓	1	✗	0
12	274.74	214.65	176.72	18.33	✓	1	✓	1	✓	1
13	269.98	212.32	180.02	15.32	✗	0	✗	0	✓	1
14	212.73	226.75	238.84	15.93	✗	0	✗	0	✓	1
15	263.03	223.05	186.51	19.65	✓	1	✓	1	✓	1

Requirements Coverage

Type of Requirement	Number of Requirements	Number of Satisfied Requirements	Number of Unmet Requirements	Percentage Covered
Function	12	10	2	83.3%
Wearability	16	14	2	87.5%
Total	28	24	4	85.7%

Was SANDRA Successful?

- All requirements needed to adequately prototype SANDRA were completed.
- SANDRA's data collection and analytical requirements were both satisfied.
- Angular data provided sufficient precision to support the rule-set for an underhand serve.

Possible Sources of Error

- Noise from sensors moving independently of the motion of the hand
- SANDRA operated under the idea that the point of impact coincides with the highest G-Force value collected during the serve, but it is possible that is incorrect
- Sampling rate causing actual point of impact to be missed

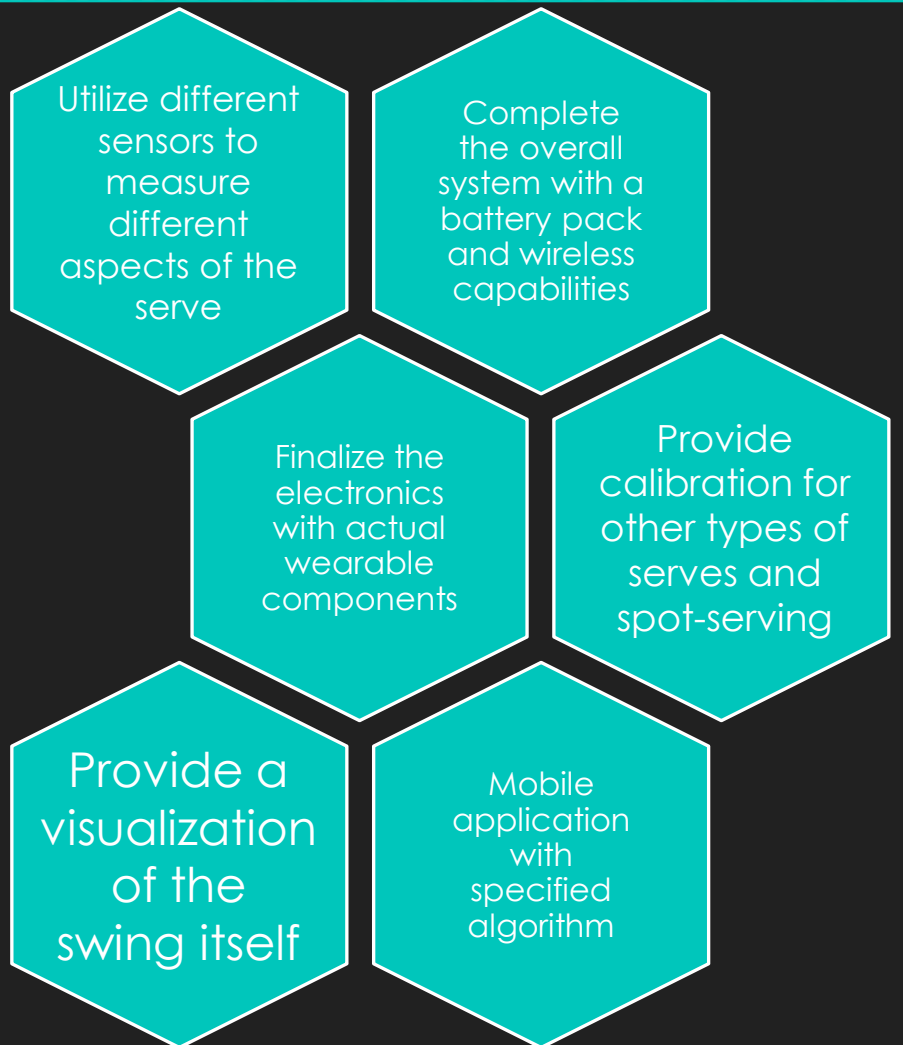
Application Highlights

SANDRA was made specifically for situations outside of the lab.

SANDRA's format makes it compatible to other sports.

SANDRA can even be used somewhere without a net.

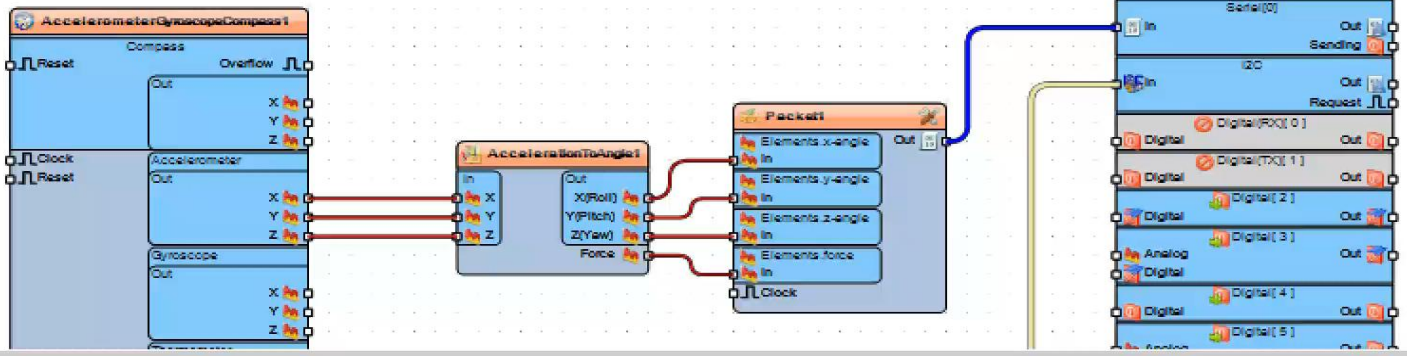
Future for SANDRA



File Edit View Arduino Help

Zoom: 70%

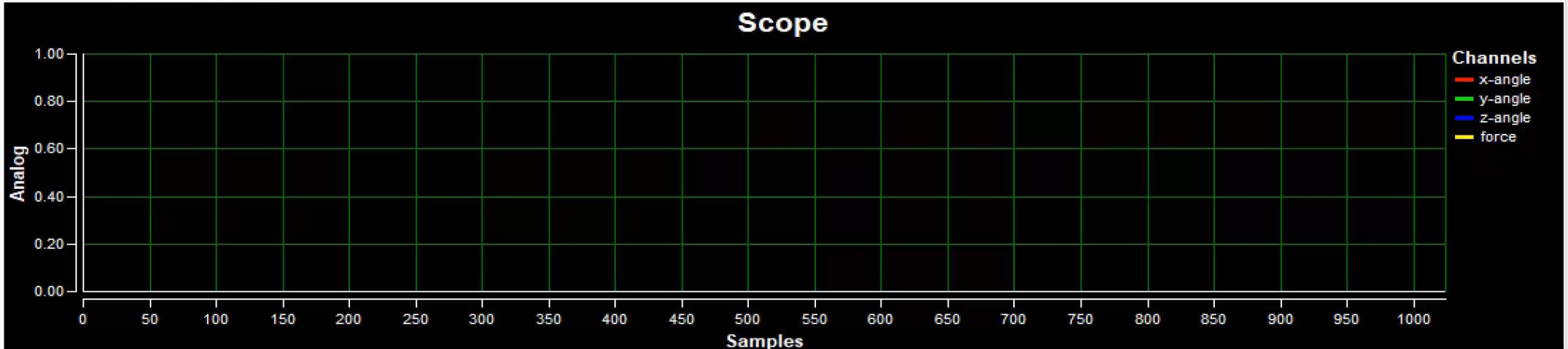




Properties

Port: COM6 Speed: 9600 Format: Packet1 Reset Log Connect

Terminal Scope Instruments



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